Algebra 1 Name____________________
Unit 5 Date____________________
Lesson 8 Practice Period__________________

For questions 1–2, solve each system of equations using substitution.

1. $\begin{cases} y = 9x - 4 \\ y = 6x + 2 \end{cases}$

2. $\begin{cases} 3x - y = 4 \\ y = \frac{1}{3}x + 4 \end{cases}$

Alexander was looking for new jobs. He got two different job offers. Job A offers Alexander an hourly position making $\$18$ an hour plus a bonus of $\$750$ a year. Job B offers Alexander an annual salary of $\$47{,}000$. The system of equations $\begin{cases} s = 18h + 750 \\ s = 47{,}000 \end{cases}$ represents the situation, where s is the salary Alexander will earn and h is the number of hours worked.

3. Solve the system of equations using substitution.

4. Interpret the solution to the system.

Yazmin is using substitution to solve the system of equations $\begin{cases} y = 7x + 10 \\ x = -y + 2 \end{cases}$. Yazmin's partial work is shown below.

$$\begin{cases} y = 7x + 10 \\ x = -y + 2 \end{cases}$$

$$y = 7(-y + 2) + 10$$
$$y = -7y + 2 + 10$$
$$y = -7y + 12$$
$$8y = 12$$
$$y = \frac{12}{8}$$

5. Yazmin made an error when solving for y. Circle the step where the mistake was made.

6. Find the correct solution for y.

7. Using the corrected y-value, find the solution to the system.

8. Yazmin substituted the second equation into the first to find y. Solve the system by substituting the first equation into the second to find x.

9. Did you get the same solution using both methods?

10. Solve the system of equations $\begin{cases} 2x - 4y = 8 \\ y = -\frac{3}{4}x + 3 \end{cases}$ by using substitution.